Online Retail — Business Analysis

Executive summary

The business recorded £9,771,318.16 in net revenue across 38 countries between 1 December 2010 and 9 December 2011, after £475,901.16 of cancellations. Net revenue is gross product sales of £10,247,219.32 plus cancellation lines. The analysis covers 19,773 original positive-sales invoices. The net-sales view holds 23,195 distinct invoice numbers because it also includes 3,422 cancellation invoices.

Five findings

- **The UK dominates net revenue in this dataset.** It produced £8,281,390.36, or 84.75% of net revenue. The Netherlands is second at £283,479.54.
- **A small group of customers carries the revenue.** 864 customers (20.0% of customers with a positive balance) generate 73.7% of their net revenue. The top ten customers generate 16.59% of identified-customer net revenue.
- **Sales are concentrated in the late-year period.** September to November produced 37.5% of net revenue across the 12 full months. November 2011 (£1,427,124.54) generated nearly three times as much net revenue as April (£482,061.93) and February (£499,464.35).
- **Cancellations are 4.64% of gross product sales, or 2.30% in a sensitivity analysis.** The sensitivity excludes invoices 581483 and 541431 (£245,653.20), which were cancelled 12 and 16 minutes after being placed. It is a comparison, not a definitive cancellation rate.
- **15.4% of net revenue has no CustomerID.** £1,505,841.23 cannot be tied to a customer, so every customer analysis covers 84.6% of net revenue.

Three recommendations

- **Protect and grow the 864 high-revenue customers.** Assign account contact, send reorder reminders before their usual purchase gap, and track their retention monthly.
- **Prepare stock and campaigns before September.** Build stock of the top five products by August and start campaigns in early September. With one year of data, confirm the pattern against a second year before committing heavily.
- **Review controls on very large orders.** Two unusually large orders worth £245,653.20 were reversed within 16 minutes of being placed. Assess whether additional confirmation or review is appropriate for high-value transactions.

1. Project overview

Objective: use SQL Server to assess retail sales performance, product and country contribution, cancellation activity, customer revenue and purchasing behaviour, and transaction timing. The source is the Online Retail transaction dataset, with invoice, product, quantity, price, date, customer and country fields.

Tools: Microsoft SQL Server. Script 01 assesses source quality; Script 02 creates prepared views; Script 03 performs business analysis.

Data preparation and metric definitions

- The raw source was retained. Exact-row deduplication reduced 541,909 source rows to 536,641 distinct rows; identical rows are treated as a documented analytical decision, not automatically proven errors.

- The deduplicated view converts invoice dates to datetime2. The source dates are interpreted in month/day/year format.
- Positive product-sales metrics use eligible non-cancellation lines with positive quantity and positive unit price, excluding special stock/service codes defined in Script 02.
- Net-sales metrics include eligible positive sales and cancellation invoice lines. Cancellation values are recorded in the month the cancellation was issued; original invoices are not comprehensively matched to cancellation invoices.
- Missing CustomerID records remain in overall net sales but are excluded from analyses that require identified customers.
- Zero-price and negative-price rows (2,516 after deduplication) and the 15 special stock codes listed below are excluded from the positive-sales and net-sales views.
- The current net-sales view contains 531,205 rows, 23,195 distinct invoice numbers, net quantity 5,291,367 and net revenue £9,771,318.16.

Rules applied

Rule	Treatment
Exact duplicates	Rows identical on all eight source columns are removed: 541,909 rows become 536,641 (5,268 rows, £21,740.98).
Special stock codes	15 non-merchandise codes are excluded from sales and cancellations (table below).
Price	Rows with a zero or negative price are excluded: 2,516 rows after deduplication, worth −£22,124.12 in total.
Cancellations	Cancellation lines remain in the net-sales view as recorded negative values. They are not automatically matched to original invoices when net revenue is calculated. Separate investigations in analyses 09 to 11 compare selected cancellation lines with earlier orders.
Missing CustomerID	135,080 raw rows (24.9%). Kept in country, product and time analyses and excluded from customer analyses.
Dates	Text dates are converted as month/day/year. The data covers 1 December 2010 to 9 December 2011.

Gross product sales are positive merchandise sales only (£10,247,219.32). Net revenue is gross product sales plus cancellation lines (£9,771,318.16). Figures in this report are labelled gross or net.

Excluded stock codes

Code	Meaning	Lines	Value
DOT	Dotcom postage	706	£206,248.77
POST	Postage	1,126	£78,101.88
M	Manual entry	317	£77,752.82
AMAZONFEE	Amazon fee	2	£13,761.09
B	Bad debt adjustment	1	£11,062.06
C2	Carriage	141	£7,051.00
BANK CHARGES	Bank charges	11	£165.00
gift_0001_10 to gift_0001_50	Gift vouchers	31	£685.81
S	Samples	2	£63.05
Total		**2,337**	**£394,891.48**

Lines and values are positive sales. The codes CRUK (charity commission) and D (discount) appear only on cancellation lines and are excluded there.

Reconciliation from raw data to net sales

Step	Rows	Value
Raw source data	541,909	£9,747,747.93
Exact duplicates removed	5,268	£21,740.98
Deduplicated data	**536,641**	**£9,726,006.95**
Positive sales before special-code exclusions	524,874	£10,642,110.80
Cancellation lines before special-code exclusions	9,251	−£893,979.73
Other lines (zero or negative price)	2,516	−£22,124.12
Special-code positive sales excluded	2,337	£394,891.48
Special-code cancellation lines excluded	583	−£418,078.57
Eligible positive merchandise sales (gross)	522,537	£10,247,219.32
Eligible cancellation lines	8,668	−£475,901.16
Net sales	**531,205**	**£9,771,318.16**

The three groups after the deduplicated total (524,874 positive, 9,251 cancellation and 2,516 other rows) add back to 536,641 rows and £9,726,006.95. Eligible positive sales plus eligible cancellation lines equal net sales.

Source data checks

Script 01 found 25,900 invoice numbers and 3,958 stock codes in the raw table. It also found 135,080 rows (24.9%) with no CustomerID, 1,454 with no description, 2,519 zero-price rows, 2 negative-price rows (−£22,124.12), 9,288 cancellation rows, and 4,879 duplicate groups containing 10,147 rows, of which 5,268 are extra copies.

2. Detailed analyses (SQL script 03)

Each analysis lists its business question, method, findings, interpretation and limitation. Figures are net unless labelled gross.

01. Top 10 products by net revenue

Business question: Which products contribute the most net revenue?

Method: Grouped eligible sales and cancellation lines by stock code and ranked products by summed line revenue; only products with positive net revenue are shown.

Findings:

- Regency Cakestand 3 Tier ranked first at £164,459.49 (net quantity 12,996).
- White Hanging Heart T-Light Holder ranked second at £99,790.93; Party Bunting ranked third at £98,243.88.
- Other leading products included Jumbo Bag Red Retrospot (£92,175.79) and Rabbit Night Light (£66,661.63).

Interpretation: These products are candidates for further inventory, seasonal demand and product-level profitability investigation.

02. Top 10 products by net quantity

Business question: Which products have the highest net unit quantities?

Method: Grouped net-sales lines by stock code and ranked by net quantity, using net revenue as a tie-breaker.

Findings:

- Small Popcorn Holder ranked first with net quantity 56,427 and net revenue £50,967.92.
- World War 2 Gliders Assorted Designs ranked second by quantity at 53,751, with £13,560.09 net revenue.
- Jumbo Bag Red Retrospot ranked third by quantity at 47,256 and generated £92,175.79 net revenue.

Interpretation: Comparing quantity with revenue distinguishes high-volume products from products with higher revenue contribution per unit.

Limitation: Quantities are net of included cancellation lines.

03. Net revenue by country

Business question: Which countries contribute the most net revenue?

Method: Summed LineRevenue by country from the net-sales view.

Findings:

- The United Kingdom led with £8,281,390.36.
- The Netherlands ranked second (£283,479.54), followed by EIRE (£259,380.02), Germany (£200,619.66) and France (£182,076.60).
- The UK contributed 84.75% of net revenue (£8,281,390.36 of £9,771,318.16).

Interpretation: The UK is the dominant market in this dataset; other markets may merit focused investigation for growth opportunities.

04. Monthly sales and cancellation trends

Business question: How does net revenue vary by month?

Method: Grouped net-sales lines by year and month, separately summing non-cancellation revenue and cancellation revenue.

Findings:

- November 2011 recorded the highest net revenue at £1,427,124.54.
- April 2011 recorded the lowest net revenue among full months at £482,061.93, followed by February 2011 at £499,464.35. February was lowest by gross sales (£507,799.87).
- December 2011 net revenue was £440,399.56, but data ends on December 9 and the month is incomplete.
- Cancellation value is assigned to the month of the cancellation transaction. April (−£33,316.06) includes one £22,998.40 invoice from customer 15749, of which £19,683.40 reverses January orders.
- September to November produced £3,500,165.50, or 37.5% of net revenue across the 12 full months (December 2010 to November 2011).

Interpretation: The increase from September through November may indicate a strong late-year sales period worth exploring for inventory and staffing planning.

Limitation: The dataset covers a limited period, so recurring seasonality cannot be established confidently.

05. Top 10 customers by net revenue

Business question: Which identified customers contribute the most net revenue?

Method: Summed net-sales revenue and quantity by non-missing CustomerID; original invoice counts count non-cancellation invoice numbers and are not cancellation-adjusted.

Findings:

- Customer 14646 ranked first (£278,778.02 across 72 original invoice IDs), followed by 18102 (£259,657.30 across 60) and 17450 (£189,575.53 across 46).
- Customer 14911 generated £128,768.24 across 198 original invoice IDs.

- Net ranking is more informative than gross ranking for this question because customers 16446 and 12346 had large matched reversals.

Interpretation: High-net-revenue customers may warrant retention attention and further behavioural analysis.

Limitation: The invoice count is not adjusted for fully cancelled original orders.

06. Top 10 customers' share of identified net revenue

Business question: How concentrated is net revenue among the top ten customers?

Method: Ranked positive-net-revenue customers and divided their combined revenue by total revenue associated with all non-missing CustomerIDs, including zero or negative customer balances in the denominator.

Findings:

- Top ten customer net revenue was £1,371,492.83.
- Total identified-customer net revenue was £8,265,476.93.
- The top ten accounted for 16.59%; the remaining identified-customer balance contributed £6,893,984.10 (83.41%).

Interpretation: The top ten form a meaningful but not majority share of identified net revenue.

Limitation: This measures revenue concentration, not customer loyalty.

07. Cancellation summary

Business question: How much cancellation activity is present in the net-sales view?

Method: Counted cancellation lines and invoice IDs and summed their quantities and line values.

Findings:

- 8,668 cancellation lines across 3,422 cancellation invoice IDs.
- 1,534 identified customers had cancellation lines.
- Total cancellation quantity was −270,017 and cancellation value was −£475,901.16.

Interpretation: Cancellation activity is material enough to monitor and investigate, especially large reversals.

Limitation: Cancellation lines may not represent independent customer returns; the data does not provide a confirmed reason for every cancellation.

08. Customers with the largest cancellation values

Business question: Which customers have the largest recorded cancellation totals?

Method: Grouped cancellation lines by CustomerID and ranked the most negative summed cancellation values first.

Findings:

- Customer 16446 had one cancellation invoice worth −£168,469.60.
- Customer 12346 had one cancellation invoice worth −£77,183.60.
- Customer 15749 had a cancellation total of −£22,998.40.

Interpretation: Large-value cancellation records are good candidates for transaction-level review.

Limitation: The ranking alone does not explain why the cancellations occurred.

09. Match large cancellations to original invoices

Business question: Do the two largest cancellation records match original invoice lines?

Method: Compared customer, stock code, quantity, unit price and timing for two invoice/cancellation pairs.

Findings:

- Invoice 541431 matched cancellation C541433 for £77,183.60; the cancellation followed 16 minutes later.
- Invoice 581483 matched cancellation C581484 for £168,469.60; the cancellation followed 12 minutes later.

- The two matched original orders totalled £245,653.20.

Interpretation: The matching line details strongly support treating these as large order reversals for sensitivity analysis.

Limitation: A matching line does not establish the business reason or prove all operational details of the reversal.

10. Cancellation-rate sensitivity with and without two large reversals

Business question: How does the recorded cancellation-to-gross-sales ratio change when the two matched reversals are removed?

Method: Compared absolute cancellation value with eligible non-cancellation gross sales, then removed £245,653.20 from each side for a sensitivity scenario.

Findings:

- Gross sales baseline: £10,247,219.32; cancellation value: −£475,901.16; ratio: 4.64%.
- After excluding the two matched reversals: gross sales £10,001,566.12; cancellation value −£230,247.96; ratio: 2.30%.

Interpretation: The two large reversals materially affect the overall ratio, so it is useful to show both values transparently.

Limitation: This is a sensitivity comparison, not a definitive operational cancellation rate; cancellations may not all be matched to their original invoices.

11. Investigate Customer 15749 cancellation

Business question: Can the cancellation invoice C550456 be linked to earlier sales lines?

Method: Compared its cancellation lines against earlier positive lines for the same customer, stock code, opposite quantity and unit price.

Findings:

- The cancellation invoice total was −£22,998.40.
- Four of the five cancellation lines, worth £19,683.40 (85.6% of the invoice), match lines from invoices 540815 and 540818, placed on 11 January 2011, 97 days earlier, with the same quantities and values.
- The line for stock code 47566B (Tea Time Party Bunting, £3,315.00) had no matching original line under the selected criteria.

Interpretation: The matches support a relationship between most lines and earlier sales and warrant documentation or further investigation.

Limitation: The matching logic cannot prove that all lines of an original order were cancelled or establish the reason for the later cancellation.

12. Original positive-sales invoice value metrics

Business question: What does an eligible original sales invoice typically generate before cancellations?

Method: Summed eligible positive product-sales lines by invoice, then calculated count, total, average, minimum and maximum invoice revenue.

Findings:

- 19,773 original positive-sales invoice IDs.
- Gross product-sales revenue: £10,247,219.32; average invoice value: approximately £518.24.
- Minimum invoice value: £0.38; maximum: £168,469.60.
- The £168,469.60 maximum is invoice 581483, which was later reversed in full (analysis 09). Excluding it and invoice 541431, the average is about £505.87 across 19,771 invoices.

Interpretation: The wide range suggests that the average should be interpreted alongside the median.

Limitation: These are pre-cancellation positive-sales values, not cancellation-adjusted order balances.

13. Median original positive-sales invoice value

Business question: What is the middle invoice value?

Method: Calculated invoice totals from eligible positive product sales and applied PERCENTILE_CONT at the 50th percentile.

Findings:

- Median invoice value was £302.20.
- The average was approximately £518.24, about £216.04 higher than the median.

Interpretation: The higher average suggests larger invoices pull the mean upward; reporting both statistics gives a fuller view.

Limitation: The distribution should be examined before attributing the difference to a particular number of outliers.

14. Average original invoice value by country

Business question: How does average original invoice value vary by country?

Method: Aggregated eligible positive product sales by country and invoice, retaining countries with at least 20 invoices.

Findings:

- The Netherlands had the highest average among included countries: about £3,052.57 across 93 invoices.
- Australia followed at about £2,466.14 across 56 invoices; Sweden was about £1,083.20 across 34 invoices.
- The UK average was about £487.37 across 17,901 invoices; France was about £482.97 and Germany about £463.61.

Interpretation: Some smaller markets have higher average invoice values and may merit further investigation.

Limitation: Country averages are affected by invoice mix and sample size.

15. Customer purchase frequency

Business question: How many identified customers are one-time versus repeat purchasers?

Method: Counted distinct original positive-sales invoice IDs for each identified customer.

Findings:

- 4,334 customers were included.
- 1,505 were one-time purchasers and 2,829 were repeat purchasers.
- Repeat customers represented 65.27% of customers in this analysis.

Interpretation: Repeat purchasing is common in this dataset and could be explored through cohort or retention analysis.

Limitation: Invoices later fully cancelled may still count because the measure is based on original positive-sales invoices.

16. Customer net-revenue segmentation

Business question: How is positive customer net revenue distributed across simple value bands?

Method: Segmented customers with positive net balances: Low below £500; Medium £500 to below £2,000; High £2,000 and above.

Findings:

- Low: 1,798 customers contributing £462,547.14.
- Medium: 1,658 customers contributing £1,708,263.82.

- High: 864 customers contributing £6,098,826.36.

Interpretation: The high segment contains fewer customers but contributes most positive customer net revenue.

Limitation: The thresholds are analytical choices, not validated business-defined customer tiers; negative and zero balances are excluded from this segmentation.

17. Customer net-balance reconciliation

Business question: How many identified customers have positive, zero or negative net balances?

Method: Grouped the summed net-sales balance for each non-missing CustomerID into balance categories.

Findings:

- Positive balances: 4,320 customers / £8,269,637.32.
- Zero balances: 12 customers / £0.00.
- Negative balances: 30 customers / −£4,160.39.
- Combined identified-customer net revenue: £8,265,476.93 across 4,362 customers.

Interpretation: This balance check explains why the positive customer balances exceed total identified-customer net revenue.

Limitation: Customer balances are based on included sales and cancellation lines only.

18. Top 3 products by net revenue in each country

Business question: Which products lead within each country?

Method: Ranked products within each country by net revenue, showing the top three positive-revenue products per country.

Findings:

- In the UK, Regency Cakestand 3 Tier led at £134,154.24, followed by White Hanging Heart T-Light Holder (£94,078.48) and Party Bunting (£92,442.63).
- In the Netherlands, Rabbit Night Light led at £9,568.48, followed by Round Snack Boxes Set of 4 Woodland (£7,991.40) and Spaceboy Lunch Box (£7,485.60).
- The SQL output contains the complete country-by-country ranking.

Interpretation: Product leaders vary by market, so country-level assortment decisions should not rely only on the overall product ranking.

Limitation: Rankings reflect net revenue, not market share within a country.

19. Sales performance by weekday

Business question: Which weekdays have the highest original positive-sales revenue?

Method: Grouped eligible positive product sales by weekday, with Sunday-first weekday numbering explicitly set in SQL.

Findings:

- Thursday led with £2,131,596.38 across 4,209 invoice IDs.
- Tuesday followed with £2,085,015.32; Wednesday recorded £1,782,348.34.
- No Saturday transactions were present in the prepared dataset.
- The two reversed orders sit inside these gross figures: £168,469.60 on a Friday (09:15, 9 December 2011) and £77,183.60 on a Tuesday (10:01, 18 January 2011). Without them Friday would be £1,598,617.66, below Monday and Wednesday, and Tuesday £2,007,831.72. Thursday would still be the highest.

Interpretation: Thursday and Tuesday were the strongest weekdays by recorded revenue in this dataset.

Limitation: The pattern reflects the dates present in the dataset and does not alone establish staffing or causal recommendations.

20. Source weekday validation

Business question: Does the full deduplicated source show the same weekday pattern?

Method: Counted deduplicated source rows and invoice IDs by weekday, including cancellation lines and special stock codes.

Findings:

- Thursday had the most source rows (103,056) and invoice IDs (5,660).
- Tuesday followed with 101,064 rows and 4,722 invoice IDs.
- Saturday had zero rows and zero invoice IDs in the deduplicated source.

Interpretation: This confirms the absence of Saturdays is present in the source, not caused by the positive-sales filter.

Limitation: The source-level counts include transaction types excluded from the positive product-sales analysis.

21. Sales performance by hour

Business question: At which hours are original positive product sales concentrated?

Method: Grouped eligible positive product sales by hour of InvoiceDateTime.

Findings:

- 12:00 had the most invoice IDs (3,205) and the highest hourly revenue (£1,412,921.32).
- 10:00 was close behind at £1,401,724.03; 13:00 generated £1,220,350.18.
- The highest activity was concentrated around late morning and early afternoon.
- The two reversed orders fall in the 09:00 hour (£168,469.60) and the 10:00 hour (£77,183.60). Without them 09:00 would be £787,240.56 and 10:00 £1,324,540.43, so 12:00 would remain the peak by a wider margin.

Interpretation: The pattern can guide further investigation of trading-hour demand.

Limitation: This is descriptive and does not establish the best opening hours without cost, staffing and coverage data.

22. Customer ID coverage

Business question: How much net sales revenue can be attributed to identified customers?

Method: Grouped the net-sales view into rows with identified and missing CustomerID.

Findings:

- Identified CustomerID: 399,656 lines and £8,265,476.93 net revenue.
- Missing CustomerID: 131,549 lines and £1,505,841.23 net revenue.
- Together, the groups reconcile to 531,205 net-sales rows and £9,771,318.16 net revenue.

Interpretation: Customer-level analysis excludes a meaningful portion of sales because the customer identifier is missing.

Limitation: Missing IDs should not be imputed without evidence; this report retains them as a separate group.

23. Overall net-sales reconciliation

Business question: What is the final size and value of the net-sales view?

Method: Summarized rows, distinct invoice numbers, net quantity, net revenue and cancellation lines from the net-sales view.

Findings:

- 531,205 rows and 23,195 distinct invoice numbers.
- Net quantity: 5,291,367; net revenue: £9,771,318.16.
- The view contains 8,668 cancellation lines.

Interpretation: This is the principal reconciliation checkpoint for the business-analysis script.

Limitation: The result depends on the deduplication, positive-price and special-code rules defined in Script 02.

3. Overall conclusions and recommended next steps

Overall conclusions

- The UK dominates net revenue in this dataset, while several smaller countries show comparatively high average original invoice values and merit separate investigation.
- Product rankings differ by measure: Regency Cakestand 3 Tier leads by net revenue, while Small Popcorn Holder leads by net quantity.
- The top ten identified customers contribute 16.59% of identified-customer net revenue. Customer 14646 has the highest current net revenue.
- Two large matched order reversals materially affect the cancellation-to-gross-sales sensitivity ratio, which falls from 4.64% to 2.30% when those reversals are excluded. This comparison depends on the stated matching assumptions.
- Repeat customers account for 65.27% of identified customers under the original positive-sales invoice definition; this is not a cancellation-adjusted retention metric.
- Missing CustomerID records contributed £1,505,841.23 in net revenue highlighting a limitation in customer-level analysis.

Recommended Next Steps

1. **Strengthen customer analysis:** Investigate the missing CustomerID records and consider cohort or retention analysis using clearly defined purchase and cancellation rules.
2. **Investigate high-value transactions and reversals:** Review the two large matched order reversals and validate the matching assumptions before drawing conclusions about cancellation patterns.
3. **Validate commercial opportunities:** Investigate high-value customers, country-level differences and leading products. Review stock readiness ahead of the late-year revenue concentration, and validate this pattern against additional years before making major inventory commitment

4. Limitations

- Revenue is not profit: product cost, shipping, operating expenses and margin are not included.
- The dataset is historical and limited in time; December 2011 is incomplete, and one period is insufficient to establish recurring seasonality.
- Cancellation-to-original-invoice matching is incomplete. Only the two largest cancellations and four of the five lines on customer 15749's invoice were matched to original orders, by query. The matching criteria do not establish operational reasons.
- The customer segments use simple revenue thresholds selected for analysis and are not confirmed business-defined categories.
- Exact-row deduplication and special-code exclusions are documented analytical choices; their impact depends on the project definitions.
- Order value, country order value, purchase frequency, weekday and hour figures use gross positive sales and still include orders that were later cancelled, including the two reversed orders.

- Weekday and hour totals are cumulative, not averages per trading day. One invoice with lines in two different hours is counted in both.
- 15.4% of net revenue (£1,505,841.23) has no CustomerID, so customer analyses cover 84.6% of net revenue.